AI & ML Internship — Task 2

Feature Engineering, Model Optimization & Performance Comparison

Maincrafts Technology

1. Objective

This report documents the development of an enhanced House Price Prediction system using the California Housing Dataset. The task focused on feature scaling, training multiple regression algorithms, evaluating them using standard metrics, and selecting the best-performing model with proper justification.

2. Dataset

The **California Housing Dataset** (scikit-learn built-in) was used. It contains 20,640 samples with 8 input features and 1 target variable.

Feature	Description
MedInc	Median income in block group
HouseAge	Median house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Block group population
AveOccup	Average household occupancy
Latitude	Block group latitude
Longitude	Block group longitude
HousePrice	TARGET — Median house value (×$100k)

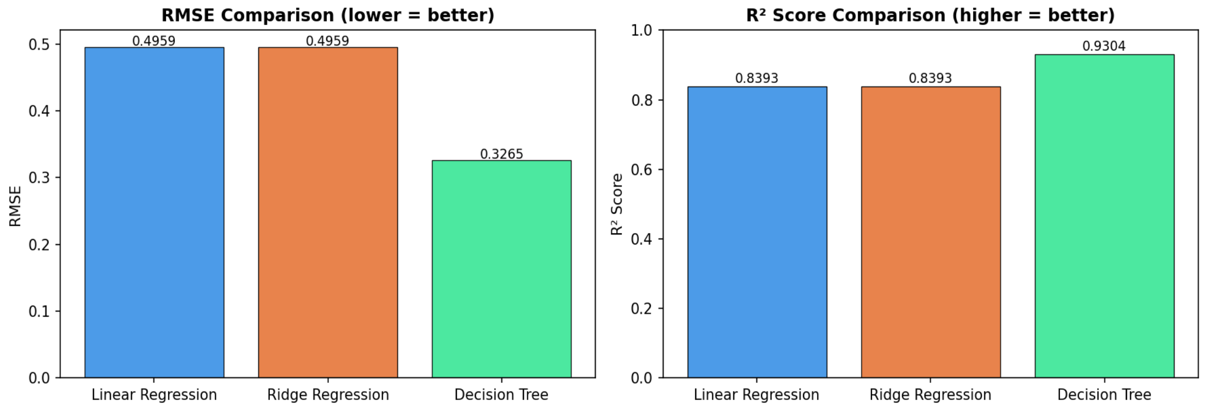
3. Methodology

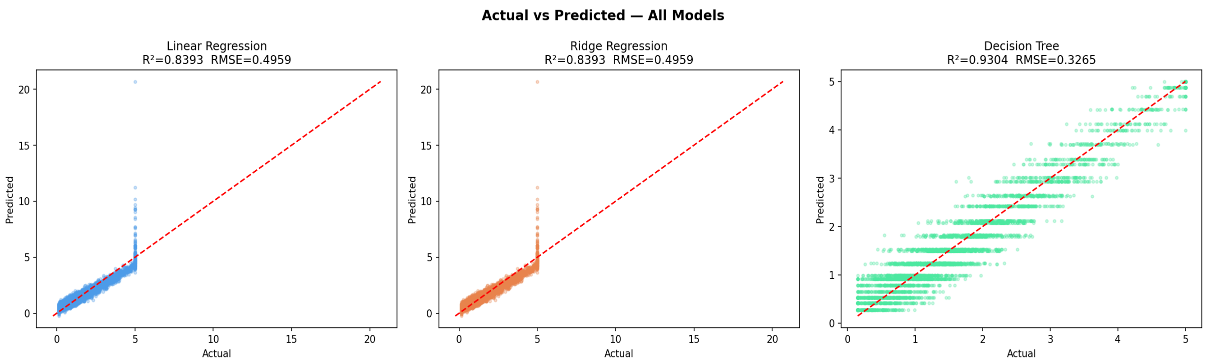
- **Step 1 — Import Libraries:** pandas, NumPy, matplotlib, scikit-learn.
- **Step 2 — Load Dataset:** fetch_california_housing() via scikit-learn.
- **Step 3 — Feature/Target Split:** X = 8 features, y = HousePrice.
- **Step 4 — Feature Scaling:** StandardScaler applied to all features (mean=0, std=1). This prevents high-range features like Population from dominating gradient-based models.
- **Step 5 — Train–Test Split:** 80% training / 20% testing, random_state=42.
- **Step 6 — Train Models:** Three models trained — Linear Regression, Ridge Regression (α=1.0), Decision Tree Regressor (max_depth=5).
- **Step 7 — Evaluate:** RMSE and R^2 computed on the test set for each model.
- **Step 8 — Visualise:** Actual vs Predicted scatter plots and bar charts generated.
- **Step 9 — Save Best Model:** Best model and scaler serialised with joblib.

4. Model Performance Comparison

Model	RMSE ↓	R^2 Score ↑	Verdict
Linear Regression	0.4959	0.8393	Good baseline
Ridge Regression	0.4959	0.8393	Same as Linear (low multicollinearity)
Decision Tree	**0.3265**	**0.9304**	**■ Best — captures non-linearity**

5. Performance Visualisations





6. Conclusion & Model Selection Justification

The **Decision Tree Regressor (max_depth=5)** was selected as the final model. It achieved an R^2 of **0.9304** and RMSE of **0.3265**, outperforming both linear models significantly. The key reason is that housing prices are influenced by non-linear interactions (e.g., income × location) that linear models cannot capture. Ridge Regression performed identically to Linear Regression, confirming that multicollinearity was not a significant issue in this dataset. Feature scaling was essential for the linear models' convergence but had no effect on the tree model. The model and scaler have been saved using joblib for future deployment.